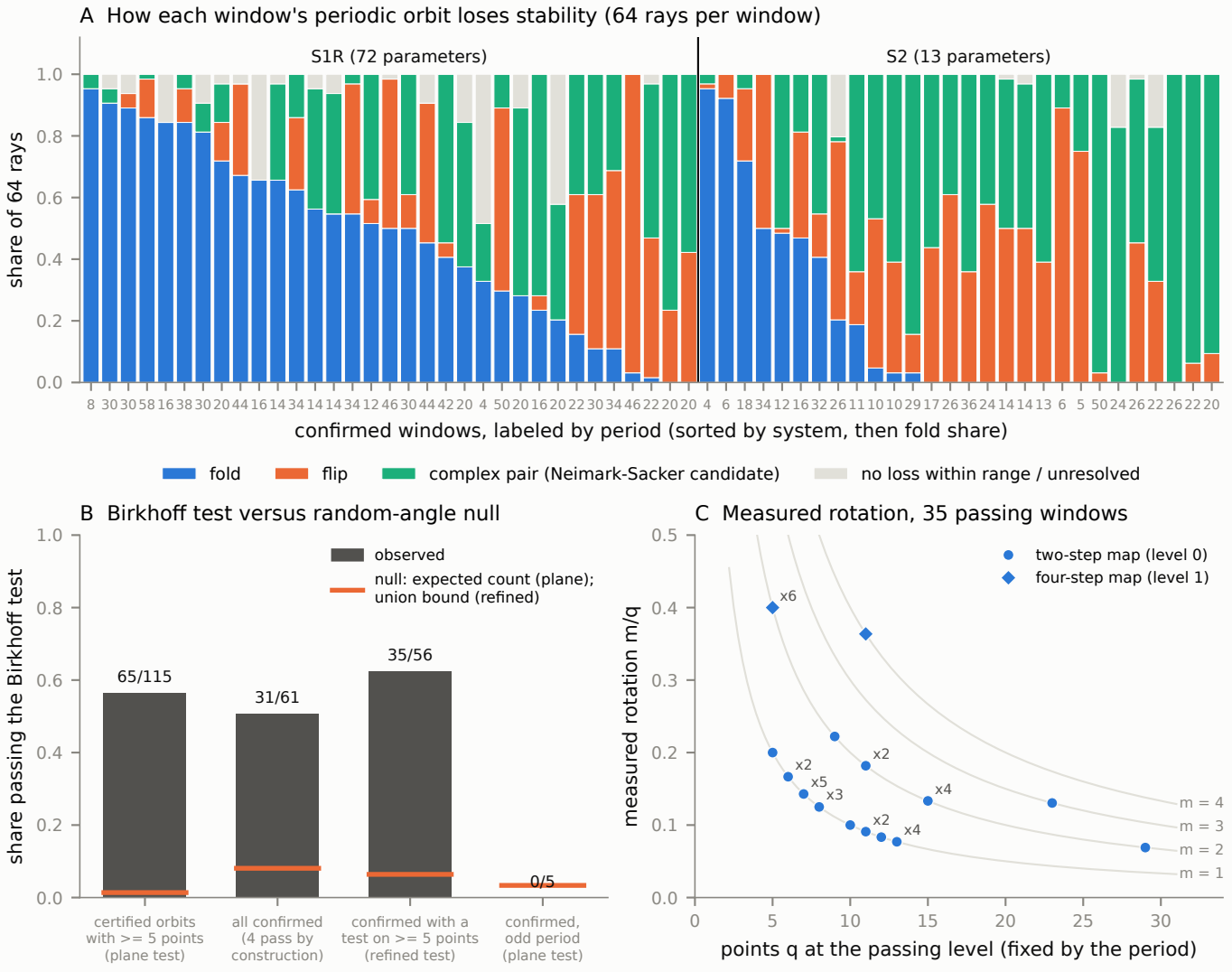
A How each window's periodic orbit loses stability (64 rays per window)
S1R (72 parameters)
S2 (13 parameters)
share of 64 rays
1.0
0.8
0.6
0.4
0.2
0.0
8 30 30 58 16 38 30 20 44 16 14 34 14 14 34 12 46 30 44 42 20 4 50 20 16 20 22 30 34 46 22 20 20 4 6 18 34 12 16 32 26 11 10 10 29 17 26 36 24 14 14 13 6 5 50 24 26 22 26 22 20
confirmed windows, labeled by period (sorted by system, then fold share)
fold
flip
complex pair (Neimark-Sacker candidate)
no loss within range / unresolved
B Birkhoff test versus random-angle null
share passing the Birkhoff test
observed
null: expected count (plane); union bound (refined)
65/115
31/61
35/56
0/5
certified orbits with >= 5 points (plane test)
all confirmed (4 pass by construction)
confirmed with a test on >= 5 points (refined test)
confirmed, odd period (plane test)
C Measured rotation, 35 passing windows
measured rotation m/q
two-step map (level 0)
four-step map (level 1)
x6
x2
x2
x5
x3
x4
x2
x4
m = 4
m = 3
m = 2
m = 1
points q at the passing level (fixed by the period)